Research scope and current progress

Dynamic Bid Prices in Revenue Management, ISE 631 IPR 2

Question groups

Theory (RQ1–3)	What does the affine value approximation retain? Why is its bound stronger? How do dual prices and column generation work?
Computation (RQ4–5)	How do size and truncation affect cost and quality? When do dynamic prices improve expected reward?
Application (RQ6)	Does the resource-allocation model transfer to surveillance missions?

Formulations and proofs	Source analysis drafted; formal audit artifact pending.
Implementations and tests	Exact DP, LPs, policies, and assertions pending.
Experiments and replication	Design proposed; no project run or published-table reproduction.

Source: Approved questions: `RESEARCH_QUESTIONS.md`; phase gates: `PLANS.md`.

Finite-horizon network revenue management

State	Period $t \in \{1, \dots, T\}$ and remaining resource vector $r \in \mathcal{R} = \prod_{i=1}^{m} \{0, \dots, c_i\}$.
Arrival	Class j arrives with probability p_{tj}; no arrival has probability $1 - \sum_j p_{tj}$. At most one request arrives per period.
Action	$u_j \in \{0, 1\}$ says whether class j would be accepted; $a_{ij} u_j \leq r_i$ for every resource i and class j.
Reward and next state	An accepted class j earns f_j and moves to $r - A_j$; otherwise capacity stays at r.

Decision principle: accept revenue now only when it covers the future value of the capacity consumed.

Source: Adelman (2007), pp. 648–649, §2.1 and eqs. (1)–(2).

Exact MDP and its linear program

Bellman recursion
$v_{T+1}(r) = 0$ and

$$v_t(r) = v_{t+1}(r) + \sum_{j: r \geq A_j} p_{tj} \left[f_j - \Delta_{t+1,j}(r) \right]_+,$$

$$\Delta_{t+1,j}(r) = v_{t+1}(r) - v_{t+1}(r - A_j).$$

For a feasible arrival, accept if

$$f_j \geq v_{t+1}(r) - v_{t+1}(r - A_j).$$

Bellman LP (D0)
Minimize $v_1(c)$ over unrestricted $v_t(r)$ subject to

$$v_t(r) \geq q_t(r, u) + \mathbb{E}[v_{t+1}(R_{t+1}) \mid r, u]$$

for every $t, r, u \in \mathcal{U}(r)$.

Its exact dual conserves probability at every time-state pair. Complementary slackness identifies optimal actions on occupied states.

Source: Adelman (2007), p. 649, eqs. (1)–(3); occupancy dual independently derived in backup slide A2.

Fluid LP and affine approximate LP

Fluid LP: expected acceptances Y_j

$$\max_{Y \geq 0} \sum_j f_j Y_j \quad \text{s.t. } AY \leq c, \quad Y_j \leq \sum_t p_{tj}.$$

Resource dual prices $\lambda_i \geq 0$ give the static test

$$f_j \geq A_j^\top \lambda.$$

This solve aggregates demand over time.

Affine value approximation

$$\hat{v}_t(r) = \theta_t + b_t^\top r.$$

Substitution in (D0) gives (D1):

$$\theta_t - \theta_{t+1} + (b_t - b_{t+1})^\top r \geq \sum_j p_{tj}(f_j - b_{t+1}^\top A_j)u_j.$$

The policy accepts a feasible class j if $f_j \geq b_{t+1}^\top A_j$.

Within one affine solve, bid prices vary by period. Re-solving can update the trajectory after capacity changes.

Source: Adelman (2007), p. 650, eqs. (4)–(6), and p. 652, eqs. (14)–(17).

Two upper-bound arguments

$$z^{LP} \;\geq\; z^{P1} \;\geq\; v_1(c)$$

1. **Affine LP versus exact MDP.** Every feasible (D1) function satisfies all Bellman inequalities. Backward induction makes it an upper bound on $v_t(r)$. Restricting the value-function family cannot lower the minimum below $v_1(c)$.
2. **Fluid LP versus affine LP.** Given feasible (P1) weights X, set $Y_j = \sum_{t,r,u} p_{tj} u_j X_{t,r,u}$. P1's mass and resource-moment equations imply $Y_j \leq \sum_t p_{tj}$ and $AY \leq c$. The two objectives agree.

Structural result: under $c_i > 0$ and $p_{tj} > 0$, a monotone optimal trajectory $b_{t,i} \geq b_{t+1,i}$ exists. An arbitrary optimal solver output need not be monotone.

Source: Adelman (2007), pp. 652–655, Theorems 1–2 and eqs. (22)–(29).

Published example and planned verification

Example 1: one resource, one seat, one class, $f = a = 1$, $T = 2$, and $p_1 = p_2 = 0.5$.

$$v_2(1) = 0.5, \qquad v_1(1) = 0.5 + 0.5(0.5) = 0.75.$$

Formulation	Value	Evidence status
Exact MDP	0.75	Hand calculation; published example
Affine LP (P1)	0.75	Published example; reproduction planned
Fluid LP	1.00	Hand calculation; published example

Verification gate: independently compare exact DP, exact LP primal/dual, fluid LP, and full affine LP on this case and a two-resource network. Test zero demand, zero capacity, terminal decisions, and infeasible arrivals.

Source: Adelman (2007), p. 653, Example 1. Project-generated solver results are pending.

Column generation and its certificate

Master: start (P1) with the feasible no-acceptance columns. Solve the restricted master and obtain θ_t, b_t.

Pricing: for each period maximize reduced profit over feasible (r, u):

$$\rho_t(r,u) = \sum_j p_{tj}(f_j - b_{t+1}^\top A_j)u_j - (b_t - b_{t+1})^\top r - \theta_t + \theta_{t+1}.$$

Add a column when its reduced profit is positive, then re-solve. Adelman formulates the period problem as an integer program.

Hand pricing check: in Example 1, zero initial dual values give reduced profit 0.5 for the period-2 accept column. Adding it raises the restricted objective from 0 to 0.5.

Stopping certificate (Proposition 3):

$$Z_\Lambda \le z^{P1} \le Z_\Lambda + \sum_{t=1}^{T} \rho_t^*, \quad \rho_t^* = \max_{r,u} \rho_t(r,u).$$

Z_Λ alone does not certify an upper bound on the exact MDP.

Source: Adelman (2007), pp. 650–651, Proposition 3; pp. 655–657, §4.4 and eqs. (33)–(36).

Proposed experiment design for IPR 3

Factors	Resources, horizon, load, fare spread, arrival timing, and single- versus multi-resource products. Start with exactly enumerable cases.
Comparators	Exact DP where tractable; fluid and certified affine bounds; fixed static and dynamic policies; separately labeled re-solving variants.
Outcomes	Bound gaps, exact or estimated policy reward, paired dynamic-minus-static difference, runtime, memory, iterations, and generated columns.
Inference	Common arrival paths and predeclared seeds; report 95% intervals for each policy mean and for paired differences. Use absolute gaps when exact value is zero.

Truncation study: record which state, column, or horizon restriction is used and whether a valid pricing certificate survives.

Source: Proposed design; no project experiment has run. Adelman (2007), pp. 657–660, supplies comparison context.

Surveillance resource allocation as a possible application

Revenue management	Candidate mission-allocation analogue
Finite resource capacity	Aircraft, sensor, or operator availability
Customer class and arrival	Mission type and uncertain request arrival
Fare and resource use	Assigned mission value and asset consumption
Bid price	Opportunity cost of committing a scarce asset

Model checks: Does capacity remain unavailable after use? Can several requests arrive together? Is utility additive and defensible? If these fail, the original MDP needs structural changes.

IPR 2 scope: conceptual mapping only. Numerical application remains outside the current project gate.

Source: Application hypothesis, not a result from Adelman (2007); project gate in `PLANS.md`.

IPR 3 gates and decisions requested

Next evidence gate	Source-linked proof notes; exact DP and LP agreement; fluid/static tests; full small affine LP; pricing versus exhaustive search; paired policy comparisons.
Professor decision 1	Is planned numerical verification sufficient for IPR 2, given that project solver evidence is not yet available?
Professor decision 2	What ALP and column-generation implementation depth is expected by IPR 3?
Professor decision 3	What level of published-table replication is expected, and which experiment factors should be fixed?
Scope confirmation	Keep the surveillance discussion conceptual until mandatory evidence and separate approval are complete.

Source: Course prompt, p. 2; project phase sequence in `PLANS.md`.

A1. Full transition and Bellman model

Let $\mathcal{U}(r) = \{u \in \{0,1\}^n : a_{ij}u_j \le r_i \ \forall i, j\}$, $p_{t0} = 1 - \sum_j p_{tj}$, and $q_t(r,u) = \sum_j p_{tj} f_j u_j$. Then

$$P_t(r' \mid r, u) = p_{t0}\mathbf{1}\{r' = r\} + \sum_j p_{tj}\mathbf{1}\{r' = r - A_j u_j\}.$$

Overlapping next states add their probabilities. With $v_{T+1} \equiv 0$,

$$v_t(r) = \max_{u \in \mathcal{U}(r)} \left\{ q_t(r,u) + \sum_{r'} P_t(r' \mid r, u) v_{t+1}(r') \right\}.$$

Initial state is c in period 1. No discount factor applies to this finite-horizon model.

Source: Adelman (2007), pp. 648–649, eq. (1); transition notation derived.

A2. Exact finite-horizon Bellman LP and occupancy dual

Primal (D0): $\min v_1(c)$, with $v_t(r) \in \mathbb{R}$, $v_{T+1} = 0$, and

$$v_t(r) - \sum_{r'} P_t(r' \mid r, u) v_{t+1}(r') \geq q_t(r, u) \quad \forall t, r \in \mathcal{R}_t, u \in \mathcal{U}(r).$$

Derived dual: $x_t(r, u) \geq 0$ and

$$\max \sum_{t,r,u} q_t(r, u) x_t(r, u)$$

subject to

$$\sum_{u} x_1(c, u) = 1, \qquad \sum_{u} x_t(r, u) = \sum_{s,u} P_{t-1}(r \mid s, u) x_{t-1}(s, u) \quad \forall t = 2, \ldots, T, \ r \in \mathcal{R}_t.$$

Strong duality gives the exact initial value. Positive $x_t(r, u)$ makes its Bellman inequality tight.

Source: Adelman (2007), p. 649, eq. (3); dual independently derived, not labeled P0 in the paper.

A3. Instructor's discounted MDP LP

For finite states S, actions $A(s)$, discount $0 \le \gamma < 1$, and initial weight $\nu \ge 0$:

$$\begin{aligned} \min_{V \in \mathbb{R}^S} \quad & \sum_s \nu(s) V(s) \\ \text{s.t.} \quad & V(s) - \gamma \sum_{s'} P(s' \mid s, a) V(s') \ge r(s, a) \quad \forall s, a \in A(s). \end{aligned}$$

The occupancy dual maximizes $\sum_{s,a} r(s,a) x(s,a)$ over $x \ge 0$ with

$$\sum_a x(s,a) - \gamma \sum_{s',a'} P(s \mid s', a') x(s', a') = \nu(s) \quad \forall s.$$

x counts ν-weighted discounted visits. Complementary slackness identifies optimal actions on occupied states. With $\nu(s) > 0$ for all s, the minimizing value vector is V^* everywhere.

Source: ISE 631 instructor prompt, p. 1; dual and full-support qualification derived.

A4. Fluid primal, dual, and policy distinction

Let $d_j = \sum_t p_{tj}$. The fluid primal is

$$\max_{Y \geq 0} \sum_j f_j Y_j \quad \text{s.t. } AY \leq c, \quad Y_j \leq d_j \ \forall j.$$

Its dual is

$$\min_{\lambda, \mu \geq 0} \ c^\top \lambda + d^\top \mu \quad \text{s.t. } A_j^\top \lambda + \mu_j \geq f_j \ \forall j.$$

Capacity shadow price λ_i is nonnegative. Complementary slackness characterizes primal-dual optima; an online rule accepting when $f_j \geq A_j^\top \lambda$ is a policy constructed from these prices, with separate performance to be evaluated.

Source: Adelman (2007), p. 650, eqs. (4)–(6) and displayed dual.

A5. Full affine Bellman LP (D1)

With $\mathcal{R}_1 = \{c\}$, $\mathcal{R}_t = \mathcal{R}$ for $t = 2, \dots, T$, and $\theta_{T+1} = 0$, $b_{T+1} = 0$, the raw affine coefficients $\theta_t, b_{t,i} \in \mathbb{R}$ are free. The program is

$$\min_{\theta, b} \quad \theta_1 + b_1^\top c$$

subject to

$$\theta_t - \theta_{t+1} + (b_t - b_{t+1})^\top r \geq \sum_j p_{tj}(f_j - b_{t+1}^\top A_j)u_j$$

$$\forall t = 1, \dots, T, \quad r \in \mathcal{R}_t, \quad u \in \mathcal{U}(r).$$

Theorem 2 permits a nonnegative monotone optimal selection only under its stated positive-capacity and positive-arrival assumptions.

Source: Adelman (2007), p. 652, eqs. (14), (16)–(17); Theorem 2, pp. 654–655.

A6. Affine dual and the bound proof

For (P1), $X_t(r,u) \geq 0$ maximizes $\sum_{t,r,u} q_t(r,u) X_t(r,u)$ subject to

$$\sum_{r,u} X_t(r,u) = 1 \quad \forall t,$$

$$\sum_{r,u} r_i X_1(r,u) = c_i, \quad \sum_{r,u} r_i X_t(r,u) = \sum_{s,u} \left(s_i - \sum_j p_{t-1,j} a_{ij} u_j \right) X_{t-1}(s,u) \quad \forall i, t > 1.$$

These are moments, not statewise flows. Set $Y_j = \sum_{t,r,u} p_{tj} u_j X_t(r,u)$. Mass gives $Y_j \leq d_j$. Telescoping resource moments gives

$$c_i = \sum_{t<T,r,u,j} p_{tj} a_{ij} u_j X_t(r,u) + \sum_{r,u} r_i X_T(r,u) \geq \sum_j a_{ij} Y_j,$$

since $r_i \geq \sum_j p_{Tj} a_{ij} u_j$. Therefore every P1 objective is fluid feasible. D1 feasibility separately gives $z^{P1} \geq v_1(c)$.

Source: Adelman (2007), pp. 652–653, eqs. (18)–(24), Theorem 1.

A7. Monotone prices: scope of Theorem 2

Under $c_i > 0$ and $p_{tj} > 0$ for every i, t, j, an optimal D1 solution exists with

$$b_{t,i} \geq b_{t+1,i} \geq 0, \qquad \theta_t \geq \theta_{t+1} \geq 0.$$

The proof uses the first period in which a resource appears in positive P1 support, complementary slackness, and changes to the dual prices that preserve feasibility and optimality.

Why "exists" matters: Example 1 also admits $\theta_2 = 0$, $b_2 = 0.5$, $\theta_1 = 0.75$, $b_1 = 0$. Its D1 objective is 0.75, but $b_1 < b_2$. Zero-demand or zero-capacity tests lie outside the theorem's stated assumptions.

Source: Adelman (2007), pp. 653–655, eqs. (25)–(29), Theorem 2 and Corollary 1; counterexample derived.

A8. Pricing, reduced costs, and stopping

For fixed $t > 1$, define $w_j = p_{tj}(f_j - b_{t+1}^{\top} A_j)$ and $d_i = b_{t,i} - b_{t+1,i}$. Adelman's pricing problem is

$$\max_{r,u} \sum_j w_j u_j - \sum_i d_i r_i - \theta_t + \theta_{t+1} \quad \text{s.t. } a_{ij} u_j \le r_i, \quad u_j \in \{0,1\}, \quad 0 \le r_i \le c_i.$$

The paper uses an integer program and generates a batch of improving columns. With global period maxima ρ_t^*, Proposition 3 yields $z^{P1} \le Z_\Lambda + \sum_t \rho_t^*$; if $Z_\Lambda > 0$, $\sum_t \rho_t^* / Z_\Lambda \le \varepsilon$ is a sufficient relative stopping rule.

Complexity caveat: A derived minimum-cut reduction for this exact pricing structure is being proof-checked. The paper itself does not claim NP-hardness or polynomial complexity.

Source: Adelman (2007), pp. 650–651, Proposition 3, and pp. 655–657, eqs. (33)–(36), Figure 2.

A9. Proposed experiment matrix

Block	Variation	Evidence to record
Verification	Example 1; two-resource network; edge cases	DP, exact LP and dual, fluid, full affine LP; values and residuals
Bound strength	Size, load, timing, fare	Exact V^* when feasible; fluid and certified affine bounds
Policy value	Fixed and re-solved policies	Tiny-case exact value; policy and paired-difference 95% intervals
Computation	Enumeration vs. columns	Runtime, memory, iterations, generated columns, pricing gaps
Truncation	Columns, states, horizon	Time saved, omitted constraints, surviving certificates

Factor levels, pilot size, final replication precision, seeds, and published replication target require approval before Phase 4 runs.

Source: Proposed design, aligned with `EXPERIMENT_SPEC.md`; no project experimental results.